Usage of GPUs at Sorbonne University

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November 2024, version 5

Contents

The goal of this tutorial is to explain how to access and use the GPU of Sorbonne University.

PPTI, the pedagogic platform of our Master (`https://www-ppti.ufr-info-p6.jussieu.fr/`), has 4 strong GPU Nvdia servers with powerfull CPUs, large memory, and disk space. You can use them remotely as described in this document. Characterics of GPUs are described in Appendix: available GPUs.

Moreover, we also have several standard PCs equiped with good Nvidia GPU (see Appendix: available GPUs). These computers have a 300 Gb storage mounted in `/tempory`. There are rebooted once per day (at 8:00).

1 Access

From your computer (`host`), you (`name`) connect to the PPTI gateway from a text terminal:

```
name@host:~$ ssh 12345678@ssh.ufr-info-p6.jussieu.fr
```

where `12345678` is your login name (and thus your student number) at the PPTI.

Then you can connect to one of the GPUs:

```
12345678@ssh:~$ ssh ppti-gpu-1
```

The other GPUs are: `ppti-gpu-2`, `ppti-gpu-3`, and `ppti-gpu-4`. They have different characteristics and powers, from `ppti-gpu-1` which is the oldest to the newest `ppti-gpu-4`.

2 Data transfer

The 4 GPUs share the same file space but it is different from the gateway and the other machines. Therefore, you must first copy the data to the gateway and then to the GPU, using the `scp` command.

- Copy from the computer to a GPU: Use the `/Vrac` folder if the data is large:

  ```
  name@host:~$ scp myfile.zip 12345678@ssh.ufr-info-p6.jussieu.fr:/Vrac
  ```

  `/Vrac` is a special folder that allows you to share data with other students. When you create a file or a folder in `/Vrac`, you are the owner (in Unix sense) of these files.

  Connect to the gateway and copy from the gateway to the GPU. You can copy directly in the home directory:

  ```
  name@host:~$ ssh 12345678@ssh.ufr-info-p6.jussieu.fr
  12345678@ssh:~$ scp /Vrac/myfile.zip ppti-gpu-1:
  ```

  Let's clean up:

  ```
  12345678@ssh:~$ rm /Vrac/myfile.zip
  ```

  We connect to the GPU and work with the transferred file:

```
12345678@ssh:~$ ssh ppti-gpu-1
12345678@ppti-gpu-1:~$ unzip myfile.zip
```

- From the gpu to our computer: Copy from the GPU to the gateway:

  ```
  name@host:~$ ssh 12345678@ssh.ufr-info-p6.jussieu.fr
  12345678@ssh:~$ scp ppti-gpu-1:myfile.zip /Vrac
  12345678@ssh:~$ Control-D (disconnect)
  ```

  On your computer, copy from the gateway:

  ```
  name@host:~$ scp 12345678@ssh.ufr-info-p6.jussieu.fr:/Vrac/monfichier.zip.
  ```

Add the `-r` option to `scp` if you want to copy directories.

To avoid having to type your passwords all the time, I recommend that you create an SSH keys pair.

Recall: on computers `ppti-*` having a good GPU, you can work in `/tempory` directory.

3 Installing Python modules:

Most of python modules usefull for machine and deep learning are already installed on the GPU.

To turn Linux/Debian to English, edit the file `~/.profile` (or create if it does not exist):

```
12345678@ppti-gpu-1:~$ nano ~/.profile
```

and add the lines:

```
source ~/.bashrc
export LANG=
```

You should restart your session to be effective.

If you want to manage your own software environnement, to get the last version of the python modules for instance, you can use `anaconda`. Type the following command:

```
12345678@ppti-gpu-1:~$ conda init bash
```

Create a python environment, it will be called ML:

```
(base) 12345678@ppti-gpu-1:~$ conda create -n ML
```

Activate the environment:

```
(base) 12345678@ppti-gpu-1:~$ conda activate ML
```

And install whatever you want:

```
(ML) 12345678@ppti-gpu-1:~$ conda install numpy pillow matplotlib jupyter
```

4 Jupyter notebook:

Remote X sessions are not recommended. But it is possible to launch a Jupiter notebook on your computer that will run remotely on the GPU. Here's how to do it.

Go on a GPU and launch the notebook (installed in the ML environment if you followed the tutorial) with the –no-brower option

```
      (ML) 12345678@ppti-gpu-1:~$ jupyter-notebook --no-browser
I 12:52:33.269 NotebookApp] Serving notebooks from local directory: /users/nfs/Enseignants/berez
I 12:52:33.269 NotebookApp] Jupyter Notebook 6.1.6 is running at:
[I 12:52:33.270 NotebookApp] http://localhost:8888/?token=efc59473f39c0bdf92eaab5f59d736ecbce74d
I 12:52:33.270 NotebookApp] or http://127.0.0.1:8888/?token=efc59473f39c0bdf92eaab5f59d736ecbce7
I 12:52:33.270 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice t
C 12:52:33.275 NotebookApp]

    To access the notebook, open this file in a browser:
        file:///users/nfs/teachers/bereziat/.local/share/jupyter/runtime/nbserver-80987-open.htm
    Or copy and paste one of these URLs:
        http://localhost:8888/?token=efc59473f39c0bdf92eaab5f59d736ecbce74d20badd1e0a
     or http://127.0.0.1:8888/?token=efc59473f39c0bdf92eaab5f59d736ecbce74d20badd1e0a
```

Note the URL of the notebook. Open a terminal on your computer and type the command:

```
(ML) 12345678@ppti-gpu-1:~$ ssh -J 12345678@ssh.ufr-info-p6.jussieu.fr -L
         8888:localhost:8888 ppti-gpu-1
```

Warning: after the option `-L` you have to use the port number that was returned by the notebook (here 8888). Open a web browser and go to the URL returned by the notebook.

5 Launching a command in "job" mode

When you disconnect from the GPU, the commands that have been run are killed by the system. If you want the command to continue to run after disconnecting from the terminal, you need to prefix it with the `nohup` command:

```
(ML) 12345678@ppti-gpu-1:~$ nohup python monscript.py
```

Another more powerful solution is to use a terminal multiplexer (`tmux` or `screen` commands).

6 Additional information

For any questions or concerns, don't hesitate to contact me. Additional information may be found here: `https://dac.lip6.fr/master/environnement-deep/`.

7 Appendix: available GPUs

Name	Model	Memory per GPU	Nb of GPUs
`ppti-gpu-1`	NVIDIA GeForce GTX TITAN	6083 MiB	1
`ppti-gpu-2`	NVIDIA A100 80GB PCIe	81920 MiB	1
`ppti-gpu-3`	Tesla P100-PCIE-12GB	12288 MiB	1
`ppti-gpu-4`	Tesla V100-SXM2-32GB	32768 MiB	4
`ppti-14-408-{01..16}`	NVIDIA GeForce RTX 4050	20470 MiB	1
`ppti-14-501-{01..16}`	NVIDIA GeForce RTX 4050	20470 MiB	1
`ppti-14-302-{01..16}`	NVIDIA GeForce RTX 3080	10240 MiB	1
`ppti-14-401-{01..16}`	NVIDIA GeForce RTX 2080	8192 MiB	1
`ppti-14-407-{01..16}`	NVIDIA GeForce RTX 2080	8192 MiB	1
`ppti-14-502-{01..16}`	NVIDIA GeForce RTX 2080	8192 MiB	1
`ppti-14-509-{01..16}`	NVIDIA GeForce RTX 2080	8192 MiB	1